

ggsplitops — Complete Documentation & KT Guide

One unique app, online + offline. This is the full knowledge-transfer (KT)

document: what the system is, how it runs, how to test, deploy, and operate it.

1. What this is

ggsplitops (display name "ggsplitops", tagline "SplitOps") is a mobile-first, collaborative expense-splitting platform for the "SplitOps / Our GG Group" circle (15 members, owner **Zubair**). Currency is **INR (₹)** throughout.

It ships in **three forms, one experience**:

Form	Entry point	Works online	Works offline
Web app (Next.js)	<code>http://localhost:3000 /</code> tunnel link / Vercel	Yes (API :3001)	Yes (localStorage state, offline-first group creation)
Android APK	<code>ggsplitops-v1.2.0.apk</code> (repo root)	Yes (WebView → live server URL, configurable by long-press)	Yes (embedded <code>assets/www/index.html</code> standalone bundle)
Backend API + DB	<code>http://localhost:3001 ,</code> Postgres :5432	n/a (it *is* the online part)	n/a

Origin: built on the open-source **SplitBills** base (MIT, credit `aaron-seq` , LICENSE kept in `ggsplitops/`), then renamed/re-themed/extended over Prompts 1–12 (full history in `PROMPTS.md`).

2. Tech stack

Layer	Technology
Monorepo	<code>pnpm 10 workspaces + Turborepo 2 (ggsplitops/)</code>

Layer	Technology
Web	Next.js 16 App Router, React 19, Tailwind CSS v4, TanStack React Query 5
API	Hono 4 (Node 22), Zod validation, Better Auth (all providers currently off)
DB	PostgreSQL 17 + pgvector, Drizzle ORM, Redis 7, MinIO (S3 stand-in), Frankfurter (self-hosted FX)
Money engine	@splitbills/core — pure TypeScript, zero I/O, tested with plain <code>node --test</code>
Android	Java WebView app (<code>com.splitops.ggsplitops</code>), hand-built with <code>aapt2/d8/zipalign/apksigner</code> , debug-signed
Infra	Docker Compose (postgres, redis, minio, frankfurter, web), Vercel (web), Cloudflare quick tunnel (public tester links)
Node	>= 22.18.0 (native TS type-stripping for core tests)

3. Repository layout

```

GACHIBOWLI-GALIDITOR/                                # GitHub repo root (this is what Vercel sees)
├─ PROMPTS.md                                           # Prompt history + change log (Prompts 1–12)
├─ DOCUMENTATION.md                                   # This file
├─ ggsplitops.apk                                       # Installable APK (mirrors in-repo build)
├─ ggsplitops-v1.1.0.apk / -v1.2.0.apk                # Versioned tester files
└─ ggsplitops/                                          # pnpm monorepo root (Vercel Root Directory)
    ├─ apps/web/                                        # Next.js 16 UI (:3000)
    │   ├─ app/ (page, login, groups, groups/[groupId], globals.css)
    │   ├─ components/ (amount, edit-expense-modal, member-manager-modal,
    │   │               month-dashboard, theme-toggle)
    │   ├─ lib/ (api, auth-client, excel-export, split-calculator, user-context)
    │   ├─ eslint.config.mjs                          # Flat config (Next 16 has no `next lint`)
    │   └─ next.config.ts                             # standalone + security headers
    ├─ apps/api/                                        # Hono API (:3001)
    │   └─ src/ (app, auth, env, index,
    │           lib/{access,errors,json,session},
    │           routes/{groups,expenses,settlements},
    │           services/{expenses,ledger})
    ├─ apps/android/                                   # Native wrapper (WebView + offline bundle)
    │   ├─ build-apk.sh                               # Full local build pipeline
    │   └─ app/src/main/{AndroidManifest.xml, res/, assets/www/index.html}
    ├─ packages/core/                                  # money.ts, split.ts, balance.ts (+ tests)
    ├─ packages/db/                                    # Drizzle schema, client, seed
    ├─ Dockerfile / docker-compose.yml / vercel.json / turbo.json
    └─ .env (git-ignored) / .env.example

```

4. Golden rules (non-negotiable, from CLAUDE.md)

1. **Money is `bigint` minor units** — never float. Amounts travel with currency codes; use `exponentOf()`, never hardcoded `100`.
 2. **Money crosses HTTP as decimal strings** (`"12.34" + currency`), never JSON numbers.
 3. **Splits go through `computeSplit`** — guarantees `sum(shares) === total` exactly.
 4. **Balances are derived, never stored** — fold over expense + settlement rows.
 5. **Every group route calls `requireMembership`** — the only row-level auth gate (plus `assertMembersInGroup` for member ids in bodies).
 6. **Never trust client-supplied split amounts** — server recomputes.
 7. **Multi-table writes use one transaction.**
 8. Domain conventions: expenses always belong to a group (1:1 = group of two); expenses reference `group_members` (null `user_id` = placeholder); soft delete via `deletedAt`; retryable mutations take `idempotencyKey`; FX snapshotted at creation.
 9. **Run `pnpm check` (typecheck + test + lint) before saying done.**
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5. Features (what the tester validates)

- **Login** — Google chooser (Praneeth account + any-email entry), 1-tap grid of all 15 members, guest-name entry; remembered on device (`ggsplitops_active_user`).
- **Groups** — SplitOps primary card, 3 curated groups (Flat 4B, Goa Trip, Cricket), custom groups; search + Favorites-Only filter + star toggles; **offline-first creation** (API failure → saved to `ggsplitops_custom_groups` with "On this device" badge, no `NetworkError`).
- **Members** — equal permissions for all; Member Manager (add/rename/remove/tag/guest); 15 seeded: Zubair (Owner), Pranu, Abhi Venkata Sai Samsani, Pavan (Kasula Pavan Sai), Prasanth, Ajay, Ajay Kumar, Dlip, Mouni, Sameena Sultana, Tharun Reddy, Uday, Devi, Hassi, Prakash.
- **Expenses** — title, amount, date, single/multi payer, participants, 4 split types (equal/exact/percentage/shares); no negatives; exact-balance validation; 2-decimal rounding; **Transport add-on** (None/Cab 🚕/Bike 🏍️ radios + fare merged into bill total, badge in history); "Last edited by <name> at <time>" on every card; search + member/date filters; delete confirm; Clear History.
- **Dashboard** — KPI cards + month calendar heatmap with working day filter; personal "My View" scoping per active account.
- **Balances & settle-up** — derived net balances, simplified transfers, one-tap Record.
- **Excel export** — `.xls` with live formulas (`=1/N`, `=ROUND`, `=SUM`, participation matrix).
- **Theme** — dark/light toggle, CSS vars via Tailwind v4 `@theme inline + [var(--color-*)]` classes (the old `[--color-*)]` shorthand emits invalid CSS — do not reintroduce).
- **Tosca automation hooks** — stable `data-testid` on login, groups, expense modal, transport radios, APK bundle (23 hooks in the APK, e.g. `apk-google-signin`, `apk-transport-cab`, `apk-`

exp-save).

6. API reference (base `http://localhost:3001/api` , session cookie required)

Method	Path	Notes
GET	<code>/health</code>	Public. {status, version, features}
GET/POST	<code>/auth/*</code>	Better Auth (all providers currently off)
GET <code>/me</code>	<code>/me</code>	Current session user
GET / POST	<code>/groups</code>	List mine / create (INR default, zod-validated)
GET	<code>/groups/:groupId</code>	Group + members
GET	<code>/groups/:groupId/balances</code>	Balances + simplified transfers
GET / POST	<code>/groups/:groupId/expenses</code>	List / create (server recomputes splits)
GET / DELETE	<code>/groups/:groupId/expenses/:expenseId</code>	Detail / delete
GET / POST	<code>/groups/:groupId/settlements</code>	List / record (idempotencyKey)

API middleware order: `logger` → `secureHeaders` → `cors(single WEB_ORIGIN)` → `csrf` → `withSession` → `requireAuth` . **Anonymous calls get 401** — by design; the web/APK degrade to on-device state instead.

7. Database

Postgres tables (`packages/db/src/schema.ts`): Better-Auth tables (`user` , `session` , `account` , `verification`) plus `friendships` , `groups` , `groupMembers` , `groupInvitations` , `categories` , `expenses` , `expensePayers` , `expenseShares` , `expenseVersions` , `recurringExpenses` , `settlements` , `attachments` , `receipts(+Items)` , `comments` , `activityLog` , `notifications` , `auditLogs` , `exchangeRates` , `expenseEmbeddings` , `aiChatMessages` , `merchantCache` . Key env: `DATABASE_URL` , `BETTER_AUTH_SECRET` (≥32 chars), `BETTER_AUTH_URL` , `WEB_ORIGIN` .

8. Runbook (daily ops)

```
cd ggsplitops
pnpm install
```

```

docker compose up -d          # postgres+redis+minio+frankfurter (use -H unix:///var
pnpm db:push                  # apply schema (dev)
pnpm dev                      # web :3000, api :3001
pnpm check                    # typecheck + test + lint (before saying done)
pnpm --filter @splitbills/core test  # 30 money-engine tests, fast, no DB

```

- **API died?** (curl :3001/api/health fails): `setsid nohup pnpm --filter @splitbills/api dev > /tmp/opencode/api.log 2>&1 < /dev/null & (plain & gets reaped — use setsid).`
- **Web stale after edits?** No volume mounts — `docker compose build web && docker compose up -d web`.
- **Public tester link?** `cloudflared tunnel --url http://localhost:3000` (quick tunnels **expire** — old links die; laptop must stay on). For permanence, use Vercel (Root Directory = `ggsplitops`).
- **APK rebuild?** `cd apps/android && bash build-apk.sh`, verify (`apksigner verify`, `apkanalyzer manifest ...`), copy to repo root + versioned filename. Bump `versionCode/versionName` in **both** `build-apk.sh` and `app/build.gradle`. Debug keystore: `apps/android/debug.keystore` (keep it — updates must reuse the same key).
- **Push?** `git add -A && git commit -m "..."` && `git push origin main` (SSH key must be registered on GitHub; see Troubleshooting).

9. Testing (app-tester role)

Tool	What it proves	Command
ESLint 9 (<code>eslint-config-next flat</code>)	0 errors (warnings ok)	<code>pnpm --filter @splitbills/web lint</code>
<code>tsc --noEmit</code> ×4 packages	Types clean	<code>pnpm typecheck</code>
<code>node --test</code> (no framework)	30/30 money tests	<code>pnpm --filter @splitbills/core test</code>
<code>next build</code>	Production build (7 routes)	<code>pnpm --filter @splitbills/web build</code>
<code>node --check</code>	APK bundle JS syntax	on extracted <code><script></code>
<code>apksigner</code> / <code>apkanalyzer</code>	APK signature + manifest truth	<code>apksigner verify</code> , <code>manifest version-code/version-name</code>
<code>curl smoke</code>	Live routes 200	<code>/</code> , <code>/groups</code> , <code>/groups/:id</code> , <code>/api/health</code>
Tosca (UI automation)	End-to-end flows	Stable <code>data-testid</code> hooks (web + APK)

Latest full pass: lint 0 errors, typecheck clean, 30/30 tests, prod build OK, APK v1.2.0 signature+manifest verified, local/tunnel/API all HTTP 200.

10. Security posture

- **Backend:** `secureHeaders` , single-origin CORS + CSRF, `requireAuth` on every `/api/*` route, Zod on all inputs, transactions on multi-table writes. **Not weakened for testers** — anonymous gets 401 by design.
- **DB:** credentials via `.env` (git-ignored, never committed); local compose uses dev defaults — **rotate to strong secrets before any production deploy**.
- **Website:** `X-Content-Type-Options` , `X-Frame-Options: DENY` , strict `Referrer-Policy` , minimal `Permissions-Policy` (verified live in response headers).
- **APK:** only `INTERNET` + `ACCESS_NETWORK_STATE` permissions; debug-signed (testers enable "Install unknown apps").

11. Troubleshooting (solved before — don't regress)

Symptom	Cause → Fix
<code>NetworkError</code> on Create Group (public link)	Browser hits <code>localhost:3001</code> off-machine + API needs session → offline-first local save (kept).
Theme unstyled / invalid CSS	<code>[--color-x]</code> Tailwind shorthand → always use <code>[var(--color-x)]</code> , <code>@theme inline</code> .
<code>pnpm lint</code> fails ("Invalid project directory.../lint")	Next 16 removed <code>next lint</code> → <code>eslint</code> . + <code>eslint.config.mjs</code> .
New <code>react-hooks/*</code> lint errors	Intentional client hydration → downgraded to <code>warn</code> in config; only <code>no-explicit-any</code> stays error.
Tunnel link dead (000)	Quick tunnels expire → start a new one, share the new URL.
<code>git push</code> → <code>Permission denied (publickey)</code>	Register <code>~/.ssh/id_ed25519.pub</code> at GitHub → SSH keys.
APK won't install (old)	Missing <code>sdk-version/aapt</code> flags, <code>API<26</code> icons, manifest screens, signing → all fixed in <code>build-apk.sh</code> /manifest/ <code>build.gradle</code> (v1.1.0+).
API 000 /dead after shell exit	Background proc reaped → restart with <code>setsid ... &</code> .
Web serves stale code	No volume mounts → <code>compose build web && up -d web</code> .
<code>pnpm db:push</code> hangs	Non-TTY prompt — run interactively or with <code>`yes \</code>

12. Release log (short)

- **v1.2.0 (code 3)** — current APK: online+offline bundle (Google sign-in, all 15 members, 4 group ledgers, cab/bike fare, balances, calendar filter, 23 testids). Files: `ggsplitops-v1.2.0.apk` ,

`ggsplitops.apk` , `ggsplitops/ggsplitops.apk` (identical).

- **v1.1.0 (code 2)** — install-fix release (sdk flags, icons, manifest, signing). File: `ggsplitops-v1.1.0.apk` .
- Web: same feature set + offline-first groups + Tosca hooks, deployed via Docker; public links via Cloudflare tunnel; Vercel needs Root Directory = `ggsplitops` .
- Full prompt-by-prompt history: `PROMPTS.md` (Prompts 1–12).

13. Glossary for KT

- **SplitOps** = the primary group/workspace ("Our GG Group"). **Curated groups** = 3 static demo groups. **On-device groups** = created while backend unreachable. **Paisa-accurate** = exact 2-decimal math. **Tosca** = UI automation tool driven via `data-testid` . **Quick tunnel** = temporary public URL (expires). **Offline-first** = UI works from localStorage, syncs when backend reachable.